# What does the quarkonium spectrum know about the heavy-quark potential?

A model-independent extraction of $V(r)$ from charmonium and bottomonium, and its consequences for $B_c$

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We determine the static heavy-quark potential $V(r)$ directly from the measured charmonium and bottomonium spectra, without assuming a functional form. $V(r)$ is represented by a cubic spline with six to twelve knots on $0.15 \le r \le 8\,\mathrm{GeV}^{-1}$ plus a Coulomb term, and is fitted together with the quark masses to the 2026 Review of Particle Physics in two ways: to 27 $J$-resolved levels with first-order spin-dependent corrections and a running coupling, and to the eleven spin-averaged multiplet centroids alone, which involves no spin-dependent input at all. The two extractions agree everywhere, and give a potential with a one-standard-deviation band: $V(r)$ is determined, relative to its value at $1\,\mathrm{GeV}^{-1}$, to 40–70 MeV for $0.7 \lesssim r \lesssim 7\,\mathrm{GeV}^{-1}$, to 0.2–0.5 GeV for $0.2 \lesssim r \lesssim 0.5\,\mathrm{GeV}^{-1}$, and not at all below $0.2\,\mathrm{GeV}^{-1}$, where the spectrum has no sensitivity and only the hyperfine splittings act. Compared at equal shape, the logarithmic and power-law potentials lie within the band; the linear rise of the Cornell potential is too steep at intermediate distance, and the Cornell fit is worse than the spline by $\Delta\chi^2 = 93$ for six extra parameters. Even with its shape free, a flavour-independent static potential does not describe the two spectra to better than $\chi^2/\mathrm{dof} \simeq 2.6$ with a 10 MeV theory error, i.e. its accuracy for spin-averaged heavy quarkonium is 15–20 MeV, with a residual pattern (charmonium $2S$–$1S$ spacing too small, bottomonium too large) that no change of $V(r)$ removes. Propagated to $c\bar{b}$, the extraction gives $B_c$ masses with uncertainties of 6–8 MeV that are independent of the choice of potential: $B_c(1S) = 6264 \pm 8$, $B_c^*(1S) = 6336 \pm 9$, $B_c(2S) = 6874 \pm 6$ MeV, and spin-averaged $1S$ and $2S$ centroids of $6315 \pm 8$ and $6900 \pm 9$ MeV against the measured $6322.9 \pm 1.3$ and $6897.8 \pm 2.1$. The $c\bar{b}$ system sits where charmonium and bottomonium put it, to the precision with which a static potential describes those two systems; the $B_c$ $2S$–$1S$ spacing, which parametric fits put 12–20 MeV above experiment, is reproduced within $0.8\sigma$ once the potential is extracted rather than assumed.

## 1 Introduction

Non-relativistic potential models describe the spectra of charmonium and bottomonium to a few tens of MeV, and have done so for fifty years with a handful of parametric forms [1–5]. The question we ask here is the inverse one: given the spectra, what is actually known about $V(r)$? Which distances are constrained, by which states, and how well? And, having let the data choose the potential, how accurate is the static-potential picture itself? The questions have a practical side. Predictions for states that have not been measured—the $P$ and $D$ waves of $B_c$, the higher bottomonia—are quoted with a "model dependence" estimated by comparing a few parametric potentials. That estimate is as arbitrary as the choice of potentials. A potential extracted from the data with a proper error band turns the model dependence into a derived quantity, and the quality of the best possible fit turns the accuracy of the static picture into a measured one.

There is a well-known answer to a related question: the static potential computed in lattice QCD [6] and the phenomenological fits agree in the range $0.2 \lesssim r \lesssim 1\,\mathrm{fm}$, i.e. $1 \lesssim r \lesssim 5\,\mathrm{GeV}^{-1}$, where the potentials of Refs. [1–3] nearly coincide despite their different functional forms [7]. What has not been done, to our knowledge, is to let the spectrum itself determine $V(r)$ with current data and to quantify the result with a covariance. That is the content of this paper.

Our method is a flexible (spline) representation of $V(r)$ fitted by least squares with a smoothness prior, which is the standard approach to ill-posed inverse problems and gives a covariance for the function. We apply it to the data set and spin-dependent treatment of a companion paper on $B_c$ [8], and separately to the spin-averaged centroids, so that the extraction can be checked against its own spin-dependent assumptions. Section 2 describes the parametrisation and the fits, Sec. 3 the extracted potential and its sensitivity, Sec. 4 the accuracy of the static picture, Sec. 5 the consequences for $B_c$, and Sec. 6 concludes.

## 2 Method

### 2.1 Flexible potential

We write

$$V(r) = -\frac{\kappa}{r} + S(r), \tag{1}$$

where $S$ is a natural cubic spline in $x = \ln r$ through $K$ knots log-spaced on $[0.15, 8]\,\mathrm{GeV}^{-1}$, the range occupied by the wave functions of the fitted states (from the inner 1% of the $b\bar{b}$ $1S$ to the outer 99% of the $c\bar{c}$ $3S$). Below the first knot $S$ is held constant, so that the Coulomb term alone sets the short-distance behaviour; beyond the last knot $S$ is continued linearly in $r$ with its slope at $8\,\mathrm{GeV}^{-1}$, a string-like extrapolation that affects only states reaching beyond the data. An additive constant in $V$ is exactly degenerate with $m_c + m_b$; we remove it by fixing the knot at $r = 1\,\mathrm{GeV}^{-1}$ so that $V(1\,\mathrm{GeV}^{-1}) = -1.166\,\mathrm{GeV}$, the

value of the Martin potential of Ref. [9]. All statements about $V(r)$ below are therefore statements about its shape relative to that point.

Bound states are obtained by integrating the radial Schrödinger equation with the Numerov method and locating eigenvalues by node-counting bisection; where fine and hyperfine structure is needed it is added in first-order perturbation theory following Eichten and Quigg [9], with the $J = L$ states of unequal-mass mesons mixed and a one-loop running coupling $\alpha_s(\mu)$, $\mu = \sqrt{m_1 m_2}$, whose scale $\Lambda$ is fitted. These are the methods of Ref. [8], where they are described in full; the $B_c$ predictions below are therefore directly comparable to the parametric ones of that paper.

### 2.2 Two data sets

*J-resolved.* The 27 charmonium and bottomonium levels below open-flavour threshold listed in Ref. [8] (Table 1 there), from the 2026 Review of Particle Physics [10]: $\eta_c$, $J/\psi$, the four $1P$ states, $\eta_c(2S)$, $\psi(2S)$, $\psi(3770)$, $\psi_2(3823)$ and $\psi_3(3842)$ for charmonium; $\eta_b$, $\Upsilon(1S)$, the four $1P$ states, $\eta_b(2S)$, $\Upsilon(2S)$, $\Upsilon_2(1D)$, the four $2P$ states, $\Upsilon(3S)$, $\chi_{b1}(3P)$ and $\chi_{b2}(3P)$ for bottomonium. The $\eta_b(2S)$ rests on a single Belle observation that the Review omits from its summary table; it enters with its full symmetrised error ($\pm 4.2$ MeV), which under the 10 MeV theory floor changes its weight by less than 5%. The free parameters are the $K-1$ knot values, $\kappa$, $m_c$, $m_b$ and $\Lambda$: eleven for $K = 8$.

*Spin-averaged.* The eleven multiplet centroids $\bar{M} = \sum_J (2J+1) M_J / \sum_J (2J+1)$ built from the same levels: $1S$, $1P$, $2S$, $1D$ for charmonium and $1S$, $1P$, $2S$, $1D$, $2P$, $3S$, $3P$ for bottomonium. Three multiplets are incomplete and are completed from the pattern of the lower ones, with an error assigned: the $b\bar{b}$ $1D$ centroid is taken as $\Upsilon_2(1D)$ ($\pm 3$ MeV; the $J = 2$ member lies within 2 MeV of the centroid in any potential model), the $3S$ centroid as $\Upsilon(3S) - \frac{3}{4}\Delta_{3S}$ with $\Delta_{3S} = 17 \pm 6$ MeV from the $1S/2S$ hyperfine pattern, and the $3P$ centroid with $\chi_{b0}(3P)$ placed $18 \pm 8$ MeV below $\chi_{b1}(3P)$ as in the $1P$ and $2P$ multiplets. Fitting centroids removes every spin-dependent assumption from the extraction; the free parameters are the knot values, $m_c$ and $m_b$ (eight for $K = 6$), with $\kappa$ held fixed since nothing in the centroid spectrum constrains it (Sec. 3).

A theory floor of 10 MeV is added in quadrature to the experimental errors of both sets.

### 2.3 Fit, prior and errors

The residual vector is the data residuals followed by the smoothness prior

$$\sqrt{\lambda}\,\frac{s_{k+1} - 2s_k + s_{k-1}}{(\Delta x)^2}, \qquad k = 2, \ldots, K-1, \quad (2)$$

on the knot values $s_k$, and the sum of squares is minimised with a trust-region least-squares algorithm. The covariance of the parameters is the inverse Gauss–Newton matrix of the full residual vector, so the prior is included in the error band: the band quantifies what the data plus the smoothness assumption determine, and we show below how much it depends on the latter. The band on $V(r)$ follows from the covariance of the knot values by linear propagation (the spline is linear in them) and vanishes at $r = 1\,\mathrm{GeV}^{-1}$ by construction.

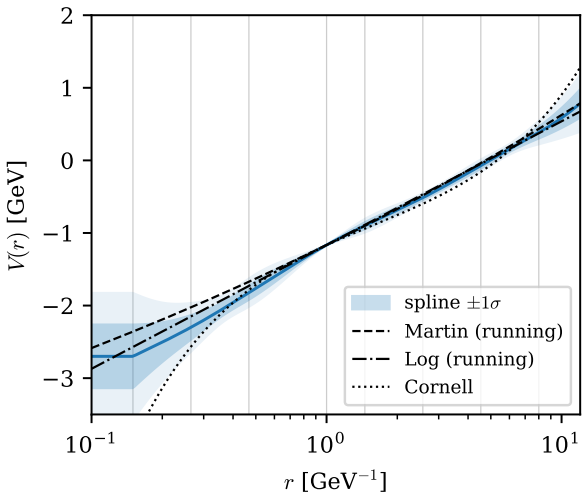



Figure 1: The potential extracted from the 27 $J$-resolved $c\bar{c} + b\bar{b}$ levels ($K = 8$, $\lambda = 1$): central value with $1\sigma$ (dark) and $2\sigma$ (light) bands, before the $\chi^2$ scaling of Sec. 2. The refitted Martin, logarithmic and Cornell potentials of Ref. [8], shifted by a constant to $V(1\,\mathrm{GeV}^{-1}) = -1.166$ GeV, are shown for comparison; vertical lines mark the knots.

Because the best fits have $\chi^2/\mathrm{dof} > 1$ (Sec. 4), all propagated errors quoted in this paper are multiplied by $\sqrt{\chi^2/\mathrm{dof}}$, as the Particle Data Group does for inconsistent inputs: 1.61 for the reference $J$-resolved fit and 1.29 for the reference centroid fit (counting the prior residuals as data). Unscaled errors would understate the uncertainty by the same factor.

The prior strength $\lambda$ and the number of knots $K$ are not fitted. We scan $K \in \{6, 8, 10, 12\}$ and $\lambda \in \{0.1, 1, 10, 100\}$, repeat the reference fits with the $c\bar{c}$ $D$ waves or the $b\bar{b}$ $3P$ (or $3S$ and $3P$) levels removed, with a 5 MeV floor, and with $\kappa$ fixed at 0.3–0.5, and quote the half-range of the predictions over these variants as a second uncertainty.

## 3 The extracted potential

Figures 1 and 2 and Table 1 show the result. The reference $J$-resolved fit ($K = 8$, $\lambda = 1$) has $\chi^2 = 41.5$ for the 27 levels with a prior contribution of 2.0; the reference centroid fit ($K = 6$, $\lambda = 1$) has $\chi^2 = 12.2$ for eleven. The two extractions, which share no spin-dependent input, agree at every distance: at $r = 3\,\mathrm{GeV}^{-1}$, $-0.404 \pm 0.047$ against $-0.396 \pm 0.044$ GeV; at $0.3\,\mathrm{GeV}^{-1}$, $-2.20 \pm 0.21$ against $-2.27 \pm 0.18$. The shape of $V(r)$ is therefore not an artefact of the perturbative spin-dependent treatment, and the fitted quark masses, $m_c = 1.805 \pm 0.024$ and

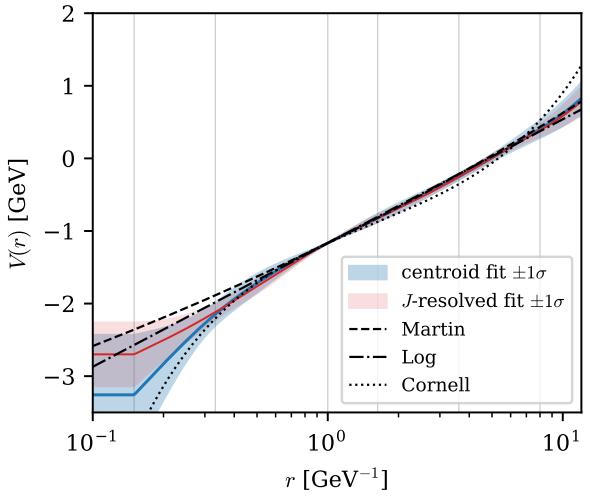



Figure 2: The same extraction from the eleven spin-averaged centroids alone ($K = 6$, $\lambda = 1$, $\kappa = 0$; blue), overlaid on the $J$-resolved band of Fig. 1 (red). The two use no common spin-dependent input.

Table 1: The extracted potential (GeV) at selected $r$ from the $J$-resolved and the centroid fit, errors scaled by $\sqrt{\chi^2/\mathrm{dof}}$, with the three parametric potentials of Ref. [8] shifted to the same value at $r = 1\,\mathrm{GeV}^{-1}$.

| $r$ | $J$-resolved | centroids | Martin | Log | Cornell |
|---|---|---|---|---|---|
| 0.2 | $-2.504 \pm 0.437$ | $-2.830 \pm 0.492$ | $-2.192$ | $-2.358$ | $-3.136$ |
| 0.3 | $-2.202 \pm 0.208$ | $-2.272 \pm 0.179$ | $-1.949$ | $-2.057$ | $-2.358$ |
| 0.5 | $-1.761 \pm 0.155$ | $-1.718 \pm 0.123$ | $-1.629$ | $-1.679$ | $-1.714$ |
| 0.75 | $-1.401 \pm 0.067$ | $-1.379 \pm 0.046$ | $-1.362$ | $-1.379$ | $-1.364$ |
| 1 | $-1.166 \pm 0.001$ | $-1.166 \pm 0.001$ | $-1.166$ | $-1.166$ | $-1.166$ |
| 1.5 | $-0.880 \pm 0.075$ | $-0.879 \pm 0.046$ | $-0.880$ | $-0.866$ | $-0.922$ |
| 2 | $-0.688 \pm 0.065$ | $-0.679 \pm 0.059$ | $-0.670$ | $-0.653$ | $-0.754$ |
| 3 | $-0.399 \pm 0.047$ | $-0.391 \pm 0.044$ | $-0.363$ | $-0.353$ | $-0.493$ |
| 4 | $-0.173 \pm 0.063$ | $-0.172 \pm 0.038$ | $-0.138$ | $-0.140$ | $-0.271$ |
| 6 | $+0.153 \pm 0.057$ | $+0.159 \pm 0.068$ | $+0.191$ | $+0.161$ | $+0.136$ |
| 8 | $+0.383 \pm 0.128$ | $+0.403 \pm 0.118$ | $+0.433$ | $+0.374$ | $+0.524$ |

$m_b = 5.180 \pm 0.023\,\mathrm{GeV}$ from the centroids, are those the spin-averaged spectrum requires.

**Fit quality versus parametric forms.** The spline is a better description of the $J$-resolved spectrum than any parametric form we have tried: the best of the three families in Ref. [8], the logarithmic potential with running coupling, has $\chi^2 = 59$ for the same data, the Martin potential 70 and the Cornell potential 134. The improvement is not an artefact of the extra freedom: $\chi^2$ is the same (40–42) for $K = 6$, 8, 10 and 12 (Table 2), so the data determine a definite shape that six knots already capture and twelve do not over-fit.

**Where the potential is known.** Figure 3 shows the band width. After scaling, $V(r)$ is determined to better than 50 MeV over $0.8 \le r \le 3.3\,\mathrm{GeV}^{-1}$ and to 50–70 MeV out to $7\,\mathrm{GeV}^{-1}$; the width grows to 0.2 GeV at $0.3\,\mathrm{GeV}^{-1}$ and 0.4 GeV at $0.2\,\mathrm{GeV}^{-1}$, and the potential is unconstrained below $0.15\,\mathrm{GeV}^{-1}$. The

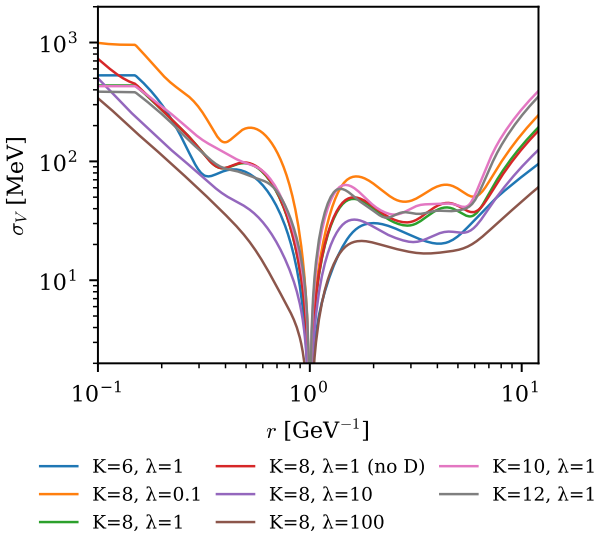



Figure 3: Width of the $1\sigma$ band (unscaled) for the $J$-resolved fits in the stability scan. The dip at $r = 1\,\mathrm{GeV}^{-1}$ is the anchoring convention.

sensitivity matrix $\partial E_i/\partial s_k$ (Fig. 4) makes the origin of this explicit. Every level responds mainly to the two or three knots nearest its mean radius: the $b\bar{b}$ $1S$ to $0.5\,\mathrm{GeV}^{-1}$, the $b\bar{b}$ $1P$ and $c\bar{c}$ $1S$ to $1.5\,\mathrm{GeV}^{-1}$, the $c\bar{c}$ $1P$ and $b\bar{b}$ $2S$, $1D$, $2P$ to $2.6\,\mathrm{GeV}^{-1}$, the $c\bar{c}$ $2S$, $D$ waves and $b\bar{b}$ $3S$, $3P$ to $4.5\,\mathrm{GeV}^{-1}$. No level has more than 3% sensitivity to the knots at 0.15 and $0.26\,\mathrm{GeV}^{-1}$. The quarkonium spectrum is blind to the Coulomb region.

**The short-distance region and the hyperfine splittings.** That blindness has a consequence for $\kappa$. In the $J$-resolved fit $\kappa$ runs to zero, and fixing it to 0.3 or 0.5 (the one-gluon-exchange value $\frac{4}{3}\alpha_s$ for $\alpha_s \simeq 0.25$–$0.4$) worsens $\chi^2$ to 47 and 51 while leaving $V(r)$ for $r > 0.5\,\mathrm{GeV}^{-1}$ unchanged within the band (Table 2); in the centroid fit $\kappa = 0$ and $\kappa = 0.4$ give the same $\chi^2$ to 0.2 (Table 3). The spectrum is not pushing $\kappa$ down—it cannot see it; the hyperfine splittings are. In first-order perturbation theory the $S$-wave hyperfine splitting is proportional to $|R(0)|^2$, which a Coulomb singularity enhances, and the fit lowers $\kappa$ to reduce the over-predicted $c\bar{c}$ splitting: even at $\kappa = 0$ the reference fit puts the $J/\psi$ 30 MeV too high and the $\eta_c$ 7 MeV too low, a $c\bar{c}$ $1S$ hyperfine splitting of 150 MeV against the measured 113, which is the largest single contribution to its $\chi^2$. This is the tension between the $c\bar{c}$ and $b\bar{b}$ hyperfine splittings found in Ref. [8] with parametric potentials, now isolated to the one region of $V(r)$ that the spin-averaged spectrum does not constrain. The short-distance form of the potential is an input, not an output, of this analysis; we carry $\kappa \in [0, 0.5]$ as a systematic.

**Comparison with parametric forms.** Because an additive constant is absorbed by the quark masses,

Table 2: Stability of the $J$-resolved extraction. For each fit: data $\chi^2$ (27 levels), prior $\chi^2$, band width $\sigma_V$ (MeV, scaled by $\sqrt{\chi^2/\mathrm{dof}}$) at four distances, and central $V$ (GeV) at three. Variants: no $D$ ($c\bar{c}$ $D$ waves removed), no $3P$ ($b\bar{b}$ $3P$ removed), floor 5 (theory floor 5 MeV), $\kappa$ fixed at 0.3 or 0.5.

| $K$ | $\lambda$ | variant | $\chi^2_{\mathrm{data}}$ | $\chi^2_{\mathrm{prior}}$ | $\sigma_V(0.3)$ | $\sigma_V(1)$ | $\sigma_V(3)$ | $\sigma_V(6)$ | $V(0.3)$ | $V(3)$ | $V(6)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 1 | – | 40.6 | 2.2 | 124 | 0 | 36 | 49 | −2.204 | −0.412 | +0.149 |
| 8 | 0.1 | – | 35.6 | 2.5 | 417 | 1 | 69 | 75 | −2.023 | −0.477 | +0.092 |
| 8 | 1 | – | 41.5 | 2.0 | 205 | 1 | 47 | 58 | −2.197 | −0.404 | +0.157 |
| 8 | 1 | floor 5 | 150.4 | 10.3 | 305 | 1 | 66 | 71 | −2.113 | −0.442 | +0.123 |
| 8 | 1 | $\kappa = 0.3$ | 46.6 | 2.1 | 244 | 1 | 50 | 61 | −2.068 | −0.402 | +0.158 |
| 8 | 1 | $\kappa = 0.5$ | 51.0 | 3.9 | 264 | 1 | 52 | 64 | −1.968 | −0.401 | +0.160 |
| 8 | 1 | no $3P$ | 31.7 | 2.4 | 224 | 1 | 52 | 114 | −2.311 | −0.407 | +0.211 |
| 8 | 1 | no $D$ | 20.8 | 1.6 | 167 | 0 | 41 | 50 | −2.235 | −0.408 | +0.137 |
| 8 | 10 | – | 46.6 | 2.3 | 133 | 0 | 36 | 48 | −2.116 | −0.370 | +0.193 |
| 8 | 100 | – | 51.3 | 7.4 | 103 | 0 | 31 | 37 | −2.017 | −0.338 | +0.220 |
| 10 | 1 | – | 40.1 | 1.7 | 267 | 1 | 67 | 89 | −2.133 | −0.362 | +0.211 |
| 12 | 1 | – | 40.4 | 1.3 | 228 | 1 | 68 | 87 | −2.131 | −0.363 | +0.201 |

Table 3: Stability of the centroid extraction (eleven spin-averaged levels). Columns as in Table 2, plus the fitted quark masses (GeV). $K = 10$ has as many parameters as data and is held only by the prior.

| $K$ | $\lambda$ | variant | $\chi^2_{\mathrm{data}}$ | $\chi^2_{\mathrm{prior}}$ | $\sigma_V(0.3)$ | $\sigma_V(3)$ | $\sigma_V(6)$ | $V(0.3)$ | $V(3)$ | $V(6)$ | $m_c$ | $m_b$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 0.1 | – | 11.9 | 0.2 | 174 | 57 | 74 | −2.318 | −0.391 | +0.164 | 1.803 | 5.178 |
| 6 | 1 | – | 12.2 | 1.2 | 177 | 44 | 68 | −2.265 | −0.396 | +0.163 | 1.805 | 5.180 |
| 6 | 1 | $\kappa = 0.2$ | 12.0 | 0.4 | 164 | 43 | 65 | −2.243 | −0.394 | +0.163 | 1.804 | 5.179 |
| 6 | 1 | $\kappa = 0.4$ | 12.0 | 0.2 | 156 | 42 | 64 | −2.214 | −0.391 | +0.163 | 1.803 | 5.178 |
| 6 | 1 | $\kappa = 0.6$ | 12.0 | 0.7 | 153 | 43 | 65 | −2.177 | −0.387 | +0.164 | 1.801 | 5.177 |
| 6 | 1 | $\kappa = 0.8$ | 12.1 | 2.0 | 155 | 45 | 67 | −2.131 | −0.383 | +0.165 | 1.799 | 5.175 |
| 6 | 1 | no $3S$, $3P$ | 6.3 | 1.1 | 163 | 39 | 104 | −2.196 | −0.394 | +0.278 | 1.791 | 5.174 |
| 6 | 1 | no $D$ | 5.2 | 1.4 | 143 | 38 | 56 | −2.310 | −0.415 | +0.132 | 1.811 | 5.188 |
| 6 | 10 | – | 15.1 | 5.4 | 183 | 44 | 68 | −2.062 | −0.355 | +0.217 | 1.782 | 5.158 |
| 8 | 1 | – | 11.3 | 1.8 | 219 | 49 | 75 | −2.332 | −0.393 | +0.172 | 1.807 | 5.182 |
| 10 | 1 | – | 11.2 | 1.6 | 212 | 56 | 82 | −2.280 | −0.370 | +0.199 | 1.794 | 5.170 |

parametric potentials must be compared at equal shape: in Figs. 1, 2 and Table 1 each has been shifted to $V(1\,\mathrm{GeV}^{-1}) = -1.166\,\mathrm{GeV}$. So compared, the Martin and logarithmic potentials lie within $1\sigma$ of both extracted bands at every $r$ between 0.3 and $8\,\mathrm{GeV}^{-1}$; their small difference in curvature is below the resolution of the data. The Cornell potential is different: its linear rise is too steep at intermediate distances, $V(3\,\mathrm{GeV}^{-1}) = -0.498\,\mathrm{GeV}$ against $-0.404 \pm 0.047$ ($J$-resolved) and $-0.396 \pm 0.044$ (centroids), $2\sigma$ in each, and the anchor-independent statement is the one above: its $\chi^2$ of 134 is worse than the spline's 41 by 93 for six extra parameters. This is the quantitative version of the long-standing observation that a linear-plus-Coulomb form fits the heavy quarkonia less well than an effective power $r^{0.1}$ or a logarithm [3, 9]: the data require the slope of the confining part to fall with distance, from $dV/dr = 0.45\,\mathrm{GeV}^2$ at $1.5\,\mathrm{GeV}^{-1}$ to $0.25\,\mathrm{GeV}^2$ at $3\,\mathrm{GeV}^{-1}$ and $0.13\,\mathrm{GeV}^2$ at $6\,\mathrm{GeV}^{-1}$, whereas a linear term has a constant slope and must compromise between the two ends.

**Stability.** Tables 2 and 3 summarise the scans. Increasing $\lambda$ from 0.1 to 10 changes $V(3\,\mathrm{GeV}^{-1})$ by 0.1 GeV and the $J$-resolved $\chi^2$ from 36 to 47; at $\lambda = 100$ the prior term ($\chi^2_{\mathrm{prior}} = 7$) is no longer negligible and the fit is over-regularised, so we exclude it. Within $0.1 \le \lambda \le 10$ the band width at $r \ge 1\,\mathrm{GeV}^{-1}$ varies by a factor of about two (the prior is doing real work there, and the band should be read with that in mind) and the central values move by less than one band width. Removing the $D$ waves or the $3P$ levels moves $V(r)$ by at most 0.02 GeV anywhere; a 5 MeV floor sharpens the unscaled band and moves nothing. The centroid extraction behaves the same way, and its fitted quark masses change by less than 0.025 GeV across all variants. The extraction is robust in the range the data cover.

# 4 How accurate is a static potential?

Letting the potential take any smooth shape answers a question that parametric fits cannot: how well can a flavour-independent static potential describe heavy quarkonium at all? The answer is in the $\chi^2$ values. With a 10 MeV theory floor the reference centroid fit has $\chi^2 = 12.2$ for four degrees of freedom, and no variant does better than $\chi^2/\mathrm{dof} \simeq 2.6$: not $\lambda = 0.1$, not

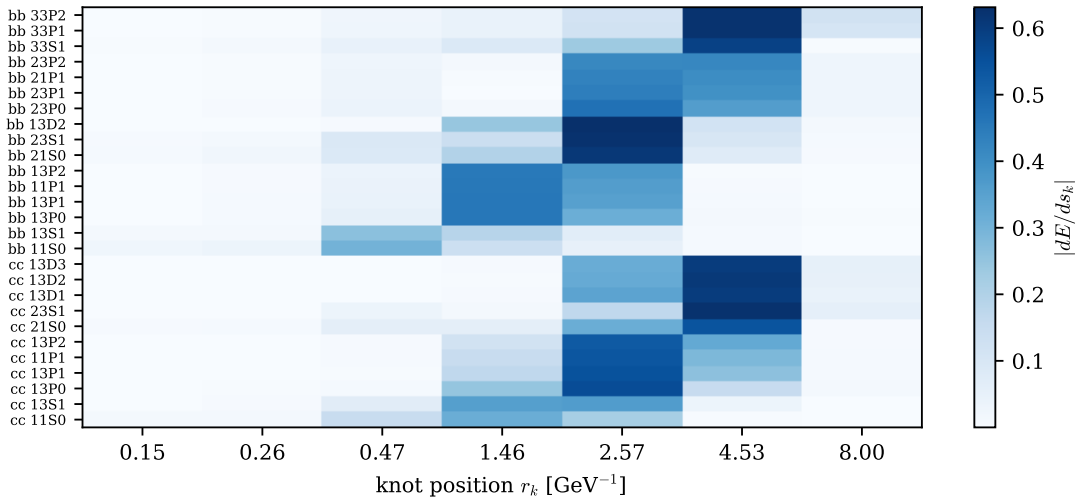



Figure 4: Sensitivity of each fitted level to the free knot values, $|\partial E_i/\partial s_k|$, for the reference $J$-resolved fit. Rows are the 27 levels (charmonium below, bottomonium above), columns the knots.

Table 4: Residuals of the reference centroid fit (MeV): the best a smooth flavour-independent static potential does for the spin-averaged spectra. Experimental errors are 0.3–5 MeV.

| level | experiment | fit | fit − exp. |
|---|---|---|---|
| $c\bar{c}$ $1S$ | 3068.7 | 3086.2 | +17.5 |
| $c\bar{c}$ $1P$ | 3525.4 | 3522.5 | −2.9 |
| $c\bar{c}$ $2S$ | 3674.0 | 3679.3 | +5.3 |
| $c\bar{c}$ $1D$ | 3822.5 | 3801.8 | −20.8 |
| $b\bar{b}$ $1S$ | 9445.0 | 9435.3 | −9.6 |
| $b\bar{b}$ $1P$ | 9899.9 | 9891.7 | −8.1 |
| $b\bar{b}$ $2S$ | 10017.3 | 10024.0 | +6.7 |
| $b\bar{b}$ $1D$ | 10163.7 | 10157.5 | −6.2 |
| $b\bar{b}$ $2P$ | 10260.2 | 10259.9 | −0.3 |
| $b\bar{b}$ $3S$ | 10342.1 | 10358.1 | +16.0 |
| $b\bar{b}$ $3P$ | 10517.2 | 10522.5 | +5.3 |

$K = 10$, where the number of parameters equals the number of data and only the prior prevents an exact fit. A floor of 17 MeV would be needed for $\chi^2/\mathrm{dof} = 1$. The $J$-resolved fit tells the same story, $\chi^2/\mathrm{dof} = 2.6$ with the same floor.

Table 4 shows what the fit gets wrong. The pattern is not noise. The $c\bar{c}$ $1S$ centroid is 17 MeV too high and the $b\bar{b}$ $1S$ 10 MeV too low, so that the $2S$–$1S$ spacing comes out 12 MeV too small for charmonium and 16 MeV too large for bottomonium; the $b\bar{b}$ $3S$ is 16 MeV too high and the $c\bar{c}$ $1D$ centroid, which contains the threshold-distorted $\psi(3770)$, 21 MeV too low. The same pattern, with the same magnitudes, appears for $K = 6$, 8 and 10 and for $\lambda = 0.1$, 1 and 10 (the $c\bar{c}$ $1S$ residual is $+17 \pm 1$ MeV in all of them), and persists when the $D$ wave or the $3S$ and $3P$ are dropped, though with the fewer remaining levels the individual residuals redistribute. A flavour-independent $V(r)$ cannot reproduce both $2S$–$1S$ spacings because they pull the potential at 1.5–2.6 GeV$^{-1}$ in opposite directions, and the data are precise enough (0.3–3 MeV) to show it.

The size of these residuals, 10–20 MeV, is what one expects from the first corrections to the static picture: the flavour-dependent $O(1/m)$ term of the heavy-quark potential [11] and the $O(v^2)$ relativistic corrections, both of which are larger for charmonium ($\langle v^2 \rangle \simeq 0.25$) than for bottomonium (0.08) and are absorbed only on average into a single $V(r)$ and two quark masses. We do not attempt to separate them here; the point is the bound. The static, flavour-independent potential describes the spin-averaged spectra of $c\bar{c}$ and $b\bar{b}$ to 15–20 MeV and no better, whatever its shape, and that number is the floor on any prediction made with it.

## 5 Consequences for $B_c$

The point of a potential with a covariance is that predictions inherit one. Table 5 gives the $B_c$ levels obtained by solving the $J$-resolved extracted potential for the $c\bar{b}$ system with the fitted $m_c$, $m_b$ and $\Lambda$, and Table 6 the spin-averaged $c\bar{b}$ centroids from the centroid extraction, each with two uncertainties: the propagated and scaled fit covariance, and the spread over the stability scan, which measures the residual dependence on $K$, $\lambda$, the floor, the data subset and $\kappa$. Every level up to the $1D$ is predicted to ±6–9 MeV, with a variant spread of 2–4 MeV; the $3S$, whose wave function extends beyond 8 GeV$^{-1}$ into the region where the potential is extrapolated, is the first level to inherit a large uncertainty (±25–30 MeV), which is as it should be. The "model dependence" of 15–35 MeV quoted in Ref. [8] from the spread of three parametric potentials was, for the lower levels, an overestimate by a factor of two to three: the spectrum determines the potential well enough that the choice of parametric form does not matter below the $3S$, and the remaining uncertainty is the accuracy of the static picture itself (Sec. 4), not the shape of $V(r)$.

Table 5: $B_c$ masses (MeV) from the $J$-resolved extraction. First error: propagated fit covariance (data plus prior), scaled by $\sqrt{\chi^2/\mathrm{dof}} = 1.61$; second: half-range over the stability-scan fits of Table 2 with $\lambda \leq 10$. Experimental values from Refs. [10, 12, 13]. Our $1P$, $1D$ are the $2P$, $3D$ of principal-quantum-number labelling.

| State | prediction | experiment |
|---|---|---|
| $1\,^1S_0$ | $6264 \pm 7 \pm 3$ | $6274.5 \pm 0.3$ |
| $1\,^3S_1$ | $6336 \pm 8 \pm 4$ | $6339.0 \pm 1.8$ |
| $2\,^1S_0$ | $6874 \pm 6 \pm 2$ | $6871.2 \pm 1.0$ |
| $2\,^3S_1$ | $6912 \pm 6 \pm 3$ | – |
| $1\,^3P_0$ | $6711 \pm 7 \pm 4$ | – |
| $1^{+}$ | $6744 \pm 6 \pm 2$ | – |
| $1^{+'}$ | $6754 \pm 6 \pm 2$ | – |
| $1\,^3P_2$ | $6767 \pm 6 \pm 4$ | – |
| $1\,^3D_1$ | $7025 \pm 6 \pm 3$ | – |
| $1\,^3D_3$ | $7026 \pm 5 \pm 3$ | – |
| $3\,^3S_1$ | $7255 \pm 27 \pm 25$ | – |
| $\Delta_{1S}$ | $71.2 \pm 6.0 \pm 6.7$ | $64.5 \pm 1.8$ |
| $\Delta_{2S}$ | $38.4 \pm 3.9 \pm 4.0$ | $35.4 \pm 2.5$ |
| $M(2\,^1S_0) - M(1\,^1S_0)$ | $609 \pm 10 \pm 3$ | $596.7 \pm 1.0$ |

Table 6: $B_c$ spin-averaged centroids (MeV) from the centroid extraction, errors as in Table 5 (scale factor 1.29; variants of Table 3 with $\lambda \leq 10$). Experimental centroids use $B_c$ and $B_c^*$ [10, 12] for the $1S$ and $B_c(2S)$ with the ATLAS+CMS $2S$ hyperfine splitting for the $2S$.

| Centroid | prediction | experiment |
|---|---|---|
| $1S$ | $6315 \pm 8 \pm 3$ | $6322.9 \pm 1.4$ |
| $2S$ | $6900 \pm 8 \pm 4$ | $6897.8 \pm 2.1$ |
| $1P$ | $6753 \pm 7 \pm 3$ | – |
| $1D$ | $7026 \pm 7 \pm 4$ | – |
| $3S$ | $7248 \pm 24 \pm 33$ | – |
| $2S - 1S$ | $585.2 \pm 12.6 \pm 4.0$ | $574.9 \pm 2.5$ |

Where experiment exists the predictions agree with it. The $B_c^*(1S)$ at $6336 \pm 9$ MeV and the $B_c(2S)$ at $6874 \pm 6$ MeV match the ATLAS [12] and PDG values; the two hyperfine splittings, $71 \pm 9$ and $38 \pm 6$ MeV, agree with the ATLAS and ATLAS+CMS values at $1\sigma$; the ground state, $6264 \pm 8$ MeV, is 10 MeV ($1.4\sigma$) below the measured 6274.5. The cleanest test is the spin-averaged one, which involves no spin-dependent input on either side: the $1S$ centroid is predicted at $6315 \pm 8 \pm 2$ against $6322.9 \pm 1.3$ MeV measured, the $2S$ at $6900 \pm 8 \pm 3$ against $6897.8 \pm 2.1$, and the $2S$–$1S$ spacing at $585 \pm 13 \pm 4$ against $574.9 \pm 2.5$ MeV: a 10 MeV excess at $0.8\sigma$.

This spacing deserves a comment, because with parametric potentials it is a persistent problem. The Cornell, logarithmic and power-law fits of Ref. [8] put the pseudoscalar $B_c$ $2S$–$1S$ spacing at 610–616 MeV against the measured $596.7 \pm 1.0$, and the $J$-resolved spline at $609 \pm 10$ (Table 5), a tension that has been read as evidence for something specific to the $c\bar{b}$ system. The centroid extraction shows that it is not. The excess is of the same size and sign as the fit's miss on the bottomonium $2S$–$1S$ spacing (+16 MeV, Table 4) and opposite to its miss on charmonium (−12 MeV); the $c\bar{b}$ system, with its reduced mass between the two, simply lands where a flavour-independent potential puts it, within the 15–20 MeV with which such a potential describes the equal-mass systems. Whatever is responsible for the $c\bar{c}/b\bar{b}$ residual pattern—the $1/m$ and $v^2$ corrections of Sec. 4—acts on $B_c$ too, at a level the current precision of the extraction cannot resolve. A $B_c$ $P$-wave measurement would not change this; what would is a lattice determination of the $O(1/m)$ potential precise enough to be included in the fit [14], which would turn the residual pattern of Table 4 into a test.

# 6 Conclusions

We have let the charmonium and bottomonium spectra determine the heavy-quark potential, with and without spin-dependent input, and found:

1. $V(r)$ is known from the spectrum, relative to its value at $1\,\mathrm{GeV}^{-1}$ and given a smoothness prior, to 40–70 MeV for $0.7 \leq r \leq 7\,\mathrm{GeV}^{-1}$ and to 0.2–0.5 GeV for $0.2 \leq r \leq 0.5\,\mathrm{GeV}^{-1}$; it is unconstrained below $0.2\,\mathrm{GeV}^{-1}$, where only the hyperfine splittings act, and above $8\,\mathrm{GeV}^{-1}$, where no bound state reaches. The extraction from spin-averaged centroids alone reproduces the one from $J$-resolved levels at every distance.

2. The extracted shape fits the 27 levels with $\chi^2 = 41$, better than any parametric form; it is consistent with a logarithmic or $r^{0.1}$ confinement, and the Cornell potential is disfavoured by $\Delta\chi^2 = 93$ because its linear rise is too steep between 2 and $5\,\mathrm{GeV}^{-1}$.

3. Even with its shape free, a flavour-independent static potential describes the spin-averaged $c\bar{c}$ and $b\bar{b}$ spectra no better than 15–20 MeV, with a definite residual pattern—the charmonium $2S$–$1S$ spacing too small, the bottomonium one too large—of the size expected for $1/m$ and relativistic corrections.

4. Propagated to $B_c$, the extraction gives predictions with ±6–9 MeV errors that do not depend on any choice of potential; the model dependence quoted for the low-lying $B_c$ levels from parametric spreads was overestimated by a factor of two to three.

5. The $B_c$ masses and hyperfine splittings, and in particular the spin-averaged $1S$ and $2S$ centroids and their spacing, agree with experiment within the scaled errors. The $2S$–$1S$ excess that every parametric fit produces is not a property of the $c\bar{b}$ system but of the static-potential description, which misses the equal-mass spacings by the same amount.

The method is general: with the same code any future measurement—a $B_c$ $P$-wave state, a $b\bar{b}$ $D$ wave, a lattice $1/m$ potential—can be added to the fit and will tighten the band in the region of $r$ to which it is sensitive, which Fig. 4 tells in advance.

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## Code and data availability

The code, all fits in the stability scans and the scripts that regenerate every table and figure are available at `https://github.com/apoppong24/bc-potential-inverse`; the experimental inputs are those of Ref. [8].

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